

TREE-01

Q10)

Question:

DS	Session	Tree 1	Question Information	Level 1 Challenge 70
Question description				
You are given a list consisting of n integers. Your task is to remove elements from the list at given positions, and report the removed elements.				
Constraints				
$1 \leq n \leq 2 \cdot 10^5$ $1 \leq x_i \leq 10^9$ $1 \leq p_i \leq n - i + 1$				
Input				
The first input line has an integer n : the initial size of the list. During the process, the elements are numbered $1, 2, \dots, k$ where k is the current size of the list.				
The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the list.				
The last line has n integers $p_1, p_2, \dots, p_n$: the positions of the elements to be removed.				

PROGRAM:

```
1  #include <stdio.h>
2
3  int bit[200005], a[200005];
4
5  void add(int i, int x, int n) {
6      while (i <= n) {
7          bit[i] += x;
8          i += i & -i;
9      }
10 }
11
12 int kth(int k, int n) {
13     int pos = 0, step = 1 << 18;
14
15     while (step) {
16         int next = pos + step;
17         if (next <= n && bit[next] < k) {
18             pos = next;
19             k -= bit[next];
20         }
21         step >>= 1;
22     }
23
24     return pos + 1;
25 }
26
27 int main() {
28     int n, i, p, pos;
29
30     scanf("%d", &n);
31
32     for (i = 1; i <= n; i++) {
33         scanf("%d", &a[i]);
34         add(i, 1, n);
35     }
36
37     for (i = 1; i <= n; i++) {
38         scanf("%d", &p);
39
40         pos = kth(p, n);
41         printf("%d", a[pos]);
42
43         add(pos, -1, n);
44
45         if (i < n)
46             printf(" ");
47     }
48
49     return 0;
50 }
```

OUTPUT:

TEST CASE 1:

```
5
2 6 1 4 2
3 1 3 1 1
1 2 2 6 4
```

TEST CASE 2:

```
6
7 2 6 1 4 2
2 2 1 2 1 1
2 6 7 4 1 2
```